

Rudra Patel

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EDUCATION

Rutgers University-New Brunswick, NJ

May 2027

Bachelor of Science in Computer Science and Data Science

PROFESSIONAL EXPERIENCE

United Window and Door

Springfield, NJ

Software Engineering Intern

Jun 2025 - Sep 2025

- Created a scalable shipping logistics tool in **Node.js** with **MSSQL** on **AWS EC2** to reduce bottlenecks in the supply chain by building **REST APIs** to allow for the dynamic serving of **100+** route updates per day.
- Developed an automated data validation pipeline using **Nodemailer** to check for and alert users via email of discrepancies in the structure of the files created, thus eliminating manual checking and saving over **25 hours** of operational overhead per week.
- Created a full-stack dynamic scheduling template system to optimize the organization of teams, in multiple functional areas, and to improve the allocation of internal resources to support enterprise-level operations.

Rutgers Business Information Technology Society (BITS)

New Brunswick, NJ

Front End Software Developer

Sep 2025 - Present

- Used **JavaScript**, **Tailwind CSS**, and **Vercel** to optimize cross-browser rendering and the overall construction of the front end of the site which increased web traffic by **60%** and reduced latency by **45%**.
- Created automated **NoSQL** data models and relational content pipelines to manage high-volume event data to eliminate **40%** of the manual effort required to maintain platforms and enhance the recruiting portals provided to our partners.

United Window and Door

Springfield, NJ

Software Engineering Intern

Jun 2024 - Sep 2024

- Developed a live data analysis dashboard employing **Next.js**, **Chart.js**, and **REST APIs** as graphical representations of clients' performance data, which resulted in the implementation of data-driven leadership decisions that led to a direct increase in revenue in Q3 of **15%**.
- Completed the redesign of enterprise internal tools and backend infrastructure using **Node**, and improved the UI/UX workflows within the company to speed up business processes by **40%** for more than **150 employees**.

PROJECTS

Real Estate Valuation Engine

- Designed an end-to-end machine learning pipeline and **Next.js** forecast system for predicting the next five years for the value of hyper-local properties and rental income.
- Created a fault-tolerant **Python ETL** ingestion layer using **SQLite** for caching; dynamically integrating temporal macroeconomic indicators and spatial demographic APIs into a low-latency **FastAPI**.
- Developed a hybrid time series forecasting model by integrating **Prophet** and **XGBoost** regressors to represent seasonal baselines, predict hyper-local residual anomalies, and create trustworthy confidence intervals.

Semantic Code Search Engine

- Engineered a privacy-first semantic search engine utilizing **Python**, **HuggingFace**, **SentenceTransformers**, and **FAISS** to enable natural language code queries on local codebase..
- Implemented advanced **Abstract Syntax Tree (AST)** parsing via Tree-sitter to chunk source code for 8+ languages in order to give live suggestions for query results using a custom **FastAPI** backend.
- Designed an automated local **MLOps** local environment to facilitate source-based indexing of source code to generate embeddings of source code and rapidly rank results based on cosine similarity to ensure that the original code is kept private.

Brain Tumor MRI Detection Tool

- Architected an end-to-end CV pipeline and custom multi-layer **CNN (Conv2D, MaxPooling, Dropout)** using the Adam optimizer to classify multi-class neurological tumors from 2D MRI scans.
- Engineered the data preprocessing workflow—including dynamic resizing, one-hot encoding, and stratified train-test splitting—maximizing feature extraction to achieve a **93.38%** peak validation accuracy.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (Postgres, MSSQL), C/C++, Java, R, HTML/CSS AI, ML & Data Science: XGBoost, Prophet, FAISS, HuggingFace, TensorFlow, Keras, scikit-learn, Pandas, NumPy

Web & Mobile Frameworks: Next.js, FastAPI, React, React Native, Node.js, Tailwind CSS, Flask Cloud, Databases & Tools: AWS (EC2), Vercel, Docker, Git, MongoDB, SQLite, Firebase, REST APIs, TravisCI